

Amanuel Asefa

Full-Stack & Mobile Software Engineer

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EXPERIENCE

Samanware

Nov 2024 – Sep 2025

Mobile App Developer

- Built an interactive e-learning mobile application featuring **live video streaming, shared learning resources, and real-time Q&A**, improving engagement between learners and instructors.
- Collaborated cross-functionally to ship features end-to-end, from UI implementation to API integration, on a fast-moving product timeline.

Sentinel Fraud Reporting Platform

Jun 2025 – Sep 2025

Full-Stack Web Developer

- Developed a community-driven fraud reporting and verification platform, combining **automated scam detection, evidence upload, and AI-driven guidance** for users.
- Implemented community alerting and moderator tooling to support trust and safety workflows at scale.

Unified Hospital Operations Platform

Dec 2025 – Mar 2026

Backend Developer

- Engineered backend systems covering the full patient lifecycle, **from reception intake through imaging and operation theatre scheduling**.
- Designed data models and APIs to coordinate multiple hospital departments within a single, reliable platform.

Sematha Technologies

Aug 2024 – Dec 2024

Full-Stack Web Developer, TADAs Sandpit Initiative

- Contributed to a multisectoral data platform integrating **AMR (antimicrobial resistance) surveillance data** from human, animal, and environmental laboratories on behalf of a national health initiative.
- Helped build reporting tools that informed public health policy and clinical decision-making.

PERSONAL PROJECTS (OPEN SOURCE)

Aegis

2025 · Open Source

CONTACT

[LinkedIn](#)

[GitHub](#)

SKILLS

Web

- MERN Stack
- LAMP Stack
- React Router v7
- Django
- Flask
- FastAPI

Mobile

- React Native
- Flutter

Core Languages

- Python
- JavaScript / TypeScript

Real-Time & Infrastructure

- WebSockets
- UDP / RTP
- Reverse Proxies & Load Balancing

LANGUAGES

- Amharic
- English
- Afan Oromo

Edge Platform & HTTP Server · github.com/manuel-spec/aegis

- Built a production-grade Node.js HTTP server from the ground up: a zero-allocation state-machine parser, a radix-tree router, a middleware pipeline, and a reverse proxy with health-checked load balancing.
- Achieved **12.4k requests/sec** with **8ms P50 latency** and 94% test coverage, alongside a live monitoring dashboard for route metrics and incidents.

Aether

2025 · Open Source

Real-Time Media SFU · github.com/manuel-spec/aether

- Designed a Selective Forwarding Unit that routes audio, video, and screen-share streams over raw UDP/RTP, including a custom congestion controller, jitter buffer, and FEC/NACK packet recovery.
- Delivered **sub-150ms end-to-end latency** at **500+ concurrent peers per room**, with graceful degradation under up to 5% packet loss.

Prism Stream

2025 · Open Source

Custom Video Codec & Streaming Framework · github.com/manuel-spec/prism-stream

- Authored a video codec from first principles, including motion estimation, DCT-based transform, adaptive quantization, and entropy coding, paired with an HLS-compatible delivery pipeline.
- Matched H.264-level visual quality at a **25% lower bitrate**, with a 5-rung adaptive bitrate ladder and a real-time encode dashboard.

CareBridge

2025 · Open Source

Developmental Assessment Platform ·

- Built a multi-platform (iOS, Android, Web) developmental assessment tool combining structured milestone tracking with **AI-driven recommendations**, a parental guidance layer, and a clinician admin portal.
- Supported 120+ tracked milestones with a fully audited record, helping reduce clinician triage time by an estimated 35%.

EDUCATION

Adama Science & Technology University (ASTU)

Apr 2021 – 2025

B.Sc. in Software Engineering, Graduated with GPA 3.41/4.00

A leading public university in Ethiopia specializing in science, engineering, and technology, with a focus on applied research and producing graduates equipped to solve real-world challenges.

Holberton ALX

Apr 2020 – Mar 2022

Backend Specialization

An intensive, project-based software engineering program focused on building scalable backend systems, with an emphasis on Python, APIs, databases, and cloud technologies through real-world, collaborative projects.